

REPORT 2

**Project Title: Service Swap (Subscription-Based
Service Exchange Platform)**

INFO 3135: Web Development Using PHP & MySQL

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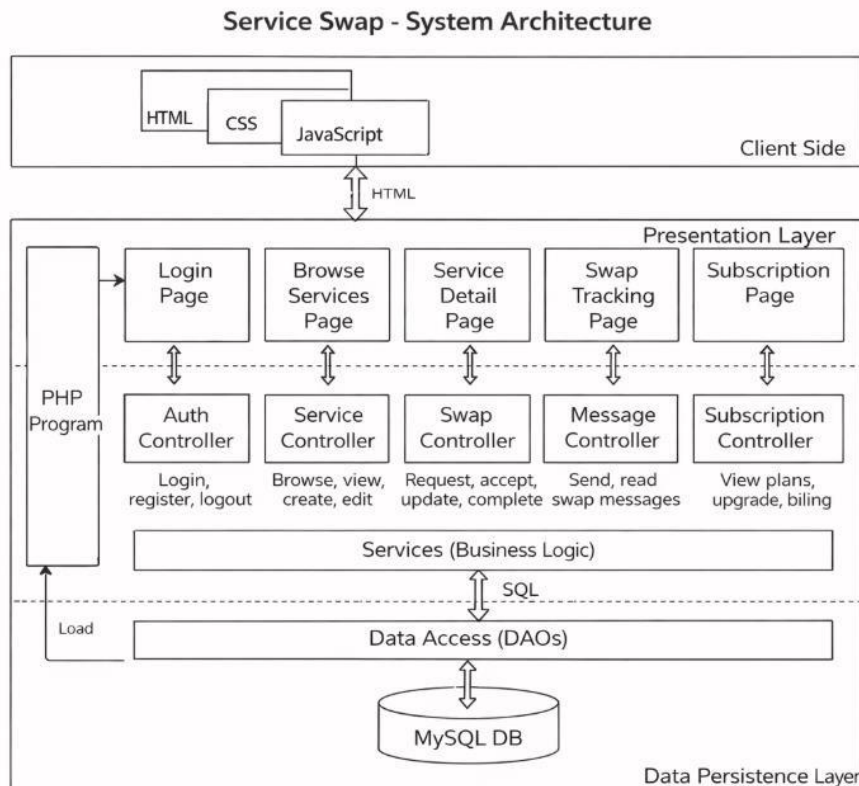
1. Introduction

This report represents the second progress stage of the Service Swap web application project. In the first progress report, the team focused on identifying the problem, defining the objectives of the platform, identifying the target users, and documenting the functional and non-functional requirements of the system.

In this phase, the project moves toward the technical design of the system. The goal is to explain how the application will be structured and how different components of the system will work together. This includes the system architecture, backend structure, business logic processing, database design, and frontend interface planning.

Service Swap is designed as a web platform that allows users to exchange services without using money. Instead, the platform uses a credit system where users earn credits by providing services and spend those credits when requesting services from other members. The system also includes messaging, ratings, dispute management, and subscription features to ensure reliability and trust among users.

This report explains the technical architecture of the platform and the diagrams that describe how the system will function during development.



2. Service System Architecture

The Service System Architecture diagram illustrates the overall structure of the Service Swap platform. The architecture is designed using a layered approach to ensure that the system is organized, scalable, and easy to maintain.

The system consists of three main layers: the presentation layer, the application layer, and the data layer.

2.1 Presentation Layer (Frontend)

The presentation layer represents the user interface of the platform. This is the part of the system that users directly interact with through their web browsers. It includes all pages such as the homepage, login page, dashboard, service listings, messaging page, and profile management page.

Technologies used in this layer include HTML for structure, CSS for styling, JavaScript for interactivity, and Bootstrap for responsive design. The presentation layer collects user input and sends requests to the backend for processing.

2.2 Application Layer (Backend Processing)

The application layer processes all user requests and applies the rules of the system. This layer is responsible for implementing the core logic of the application. It validates user input, manages sessions, processes service requests, and ensures that the correct system workflow is followed.

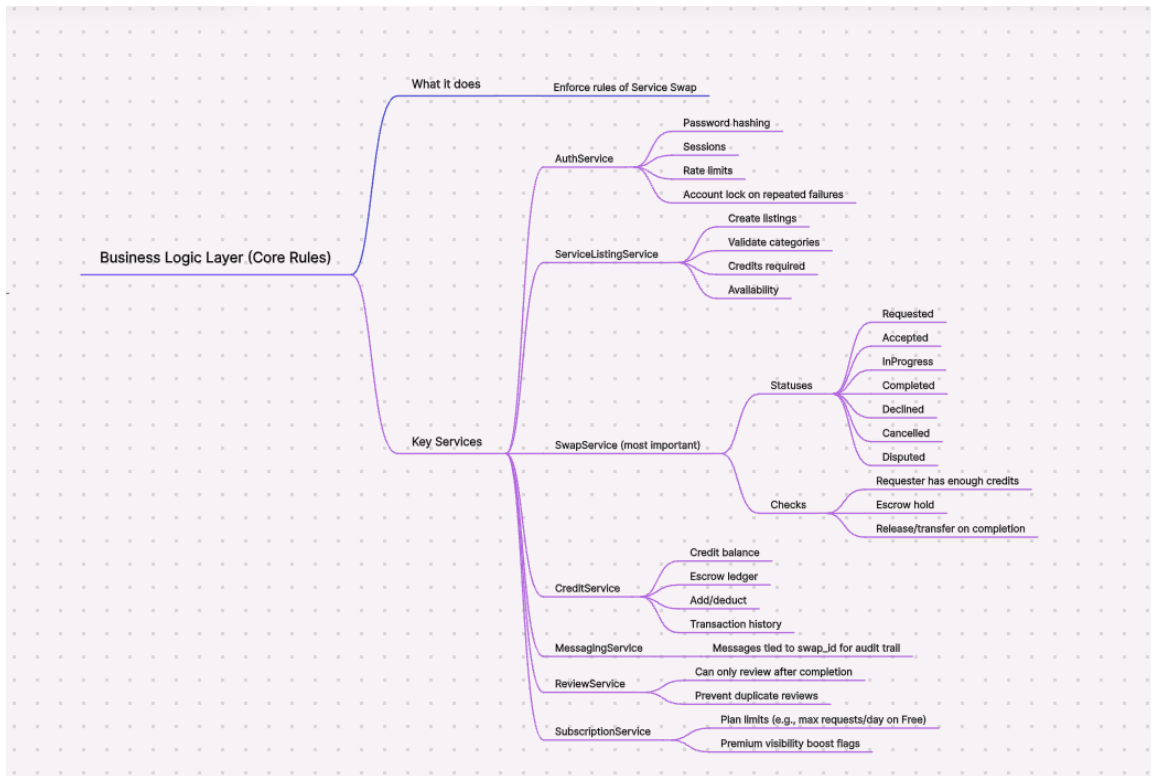
This layer will be implemented using PHP, which communicates with the database to retrieve and store information.

2.3 Data Layer (Database)

The data layer is responsible for storing all system data in the MySQL database. It contains all information related to users, services, transactions, messages, and reviews. The database ensures that data is securely stored and can be efficiently retrieved when required by the application layer.

3. Business Logic Layer

The Business Logic Layer is responsible for handling the internal processing of the platform. It ensures that all system operations follow the defined rules and workflows.



3.1 User Authentication Logic

The system validates login credentials when a user attempts to access the platform. Passwords are encrypted and stored securely. Sessions are created after successful login to maintain authentication throughout the user's interaction with the platform.

3.2 Service Management Logic

Users are able to create, edit, and delete service listings. Each service listing contains details such as service title, description, category, credit value, and availability. The system verifies that only the service owner can modify their listing.

3.3 Service Swap Processing

When a user requests a service, the system generates a swap request. The provider receives the request and has the option to accept or decline it. If accepted, the swap becomes active and both users can communicate through messaging.

3.4 Credit Management

The credit system ensures fairness in service exchanges. When a request is made, the required credits are temporarily locked in escrow. After the service is completed and confirmed, the credits are transferred to the provider.

3.5 Ratings and Trust System

After each completed service exchange, both users can leave ratings and reviews. These reviews help other users determine the reliability and quality of service providers.

3.6 Dispute Resolution

If any issue occurs during a service exchange, users can submit a dispute report. The dispute will then be reviewed by the system administrator, who can investigate the issue and apply appropriate actions.

4. Backend Architecture

The backend architecture describes how server-side components are structured and how the application logic is implemented.

The backend is developed using PHP and communicates with the MySQL database.

The architecture is divided into several modules:

Authentication Module - manages user registration, login, logout, and password encryption.

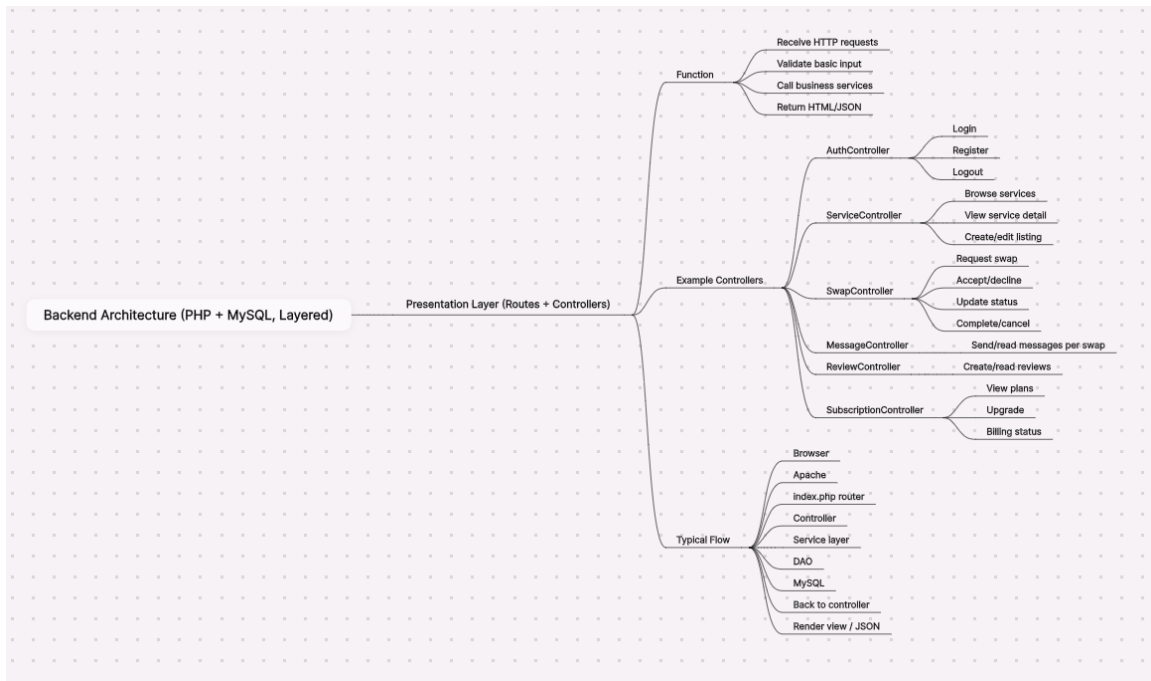
Service Module - handles creation, editing, deletion, and display of service listings.

Swap Request Module - manages the process of requesting services, accepting requests, tracking swap status, and confirming completion.

Messaging Module - allows users to communicate with each other during a service exchange.

Review Module - stores ratings and feedback provided after a swap is completed.

Admin Module - allows administrators to monitor users, manage disputes, and control system operations.



5. Database Design

The database design defines how information will be structured and stored in the MySQL database.

Users Table - stores user information including user ID, name, email, password, skills, profile description, and subscription status.

Services Table - stores service listings created by users including service title, description, category, and credit value.

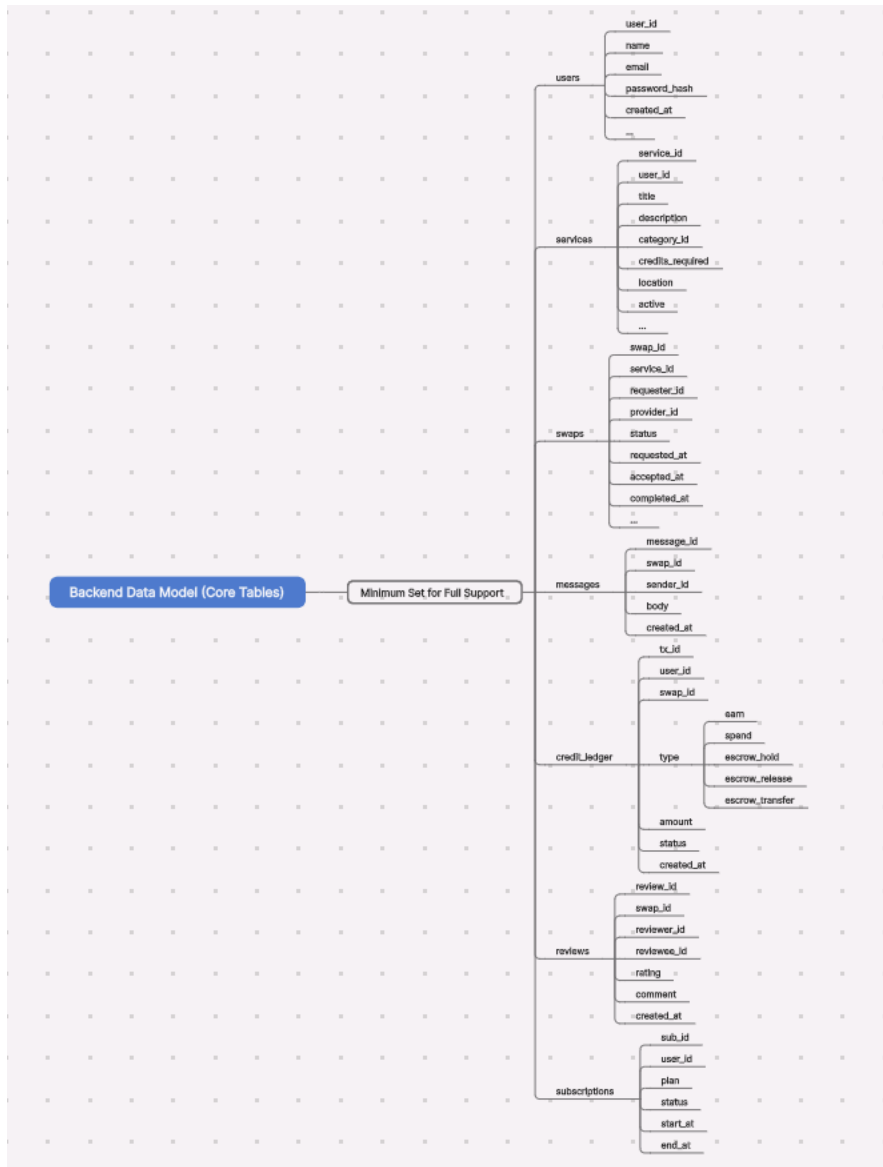
Swap Requests Table - records service exchange requests between users and tracks their status.

Messages Table - stores communication between users during service exchanges.

Reviews Table - records ratings and feedback submitted after service completion.

Disputes Table - records complaints or issues reported by users which require administrative review.

This structured database design ensures data integrity and efficient retrieval of information.



6. Use Case Diagram Explanation

The Use Case Diagram represents the interactions between users and the Service Swap system.

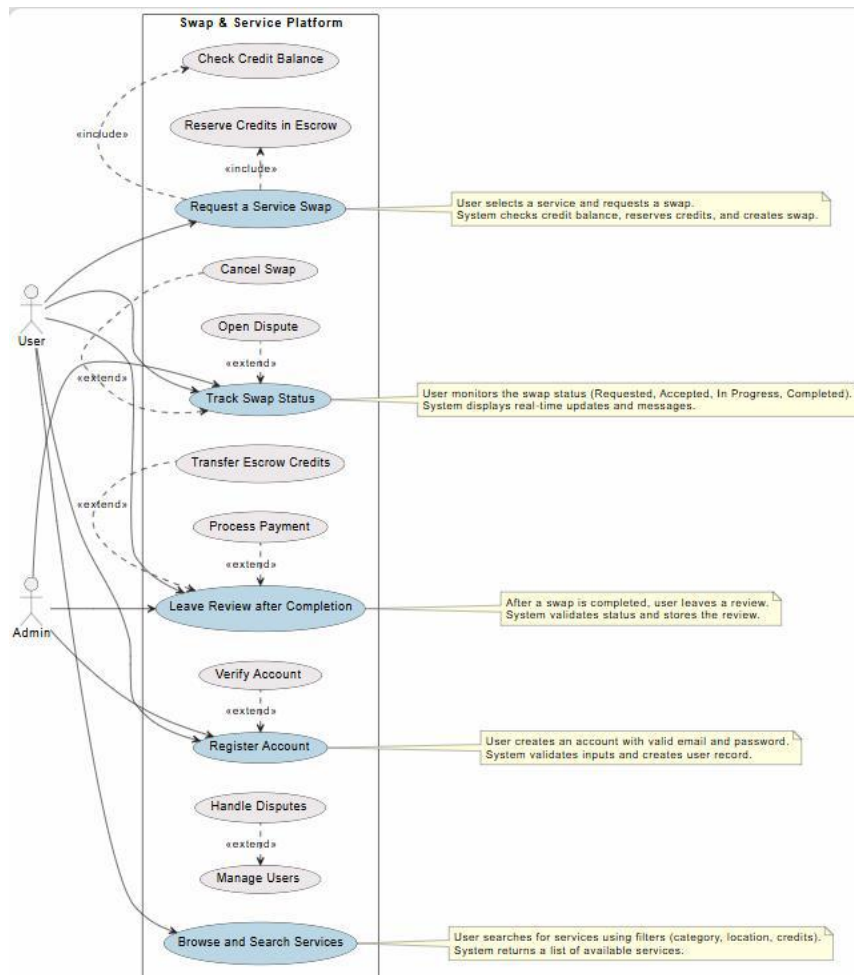
Member (User)

Members can perform several activities including registering an account, logging into the platform, managing their profile, browsing available services, requesting services, communicating with service providers, confirming service completion, and leaving ratings and reviews.

Administrator

The administrator is responsible for managing the platform operations. This includes monitoring user activity, resolving disputes, moderating service listings, and managing subscription plans.

The use case diagram visually represents these interactions and helps developers understand how different system components interact with users.



7. Frontend Wireframes

Frontend wireframes show the layout of the system interface before the final design is implemented.

Home Page - introduces the platform and allows users to browse services or sign up.

User Dashboard - shows active swaps, available credits, and quick access to user features.

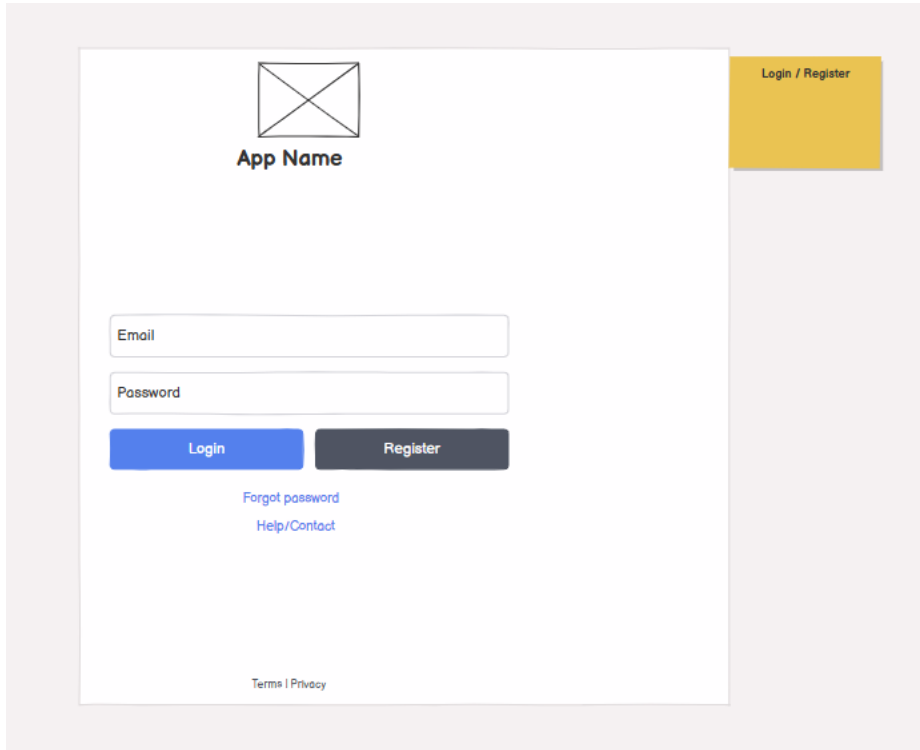
Service Listings Page - displays services with filtering and search options.

Service Detail Page - provides detailed information about a specific service.

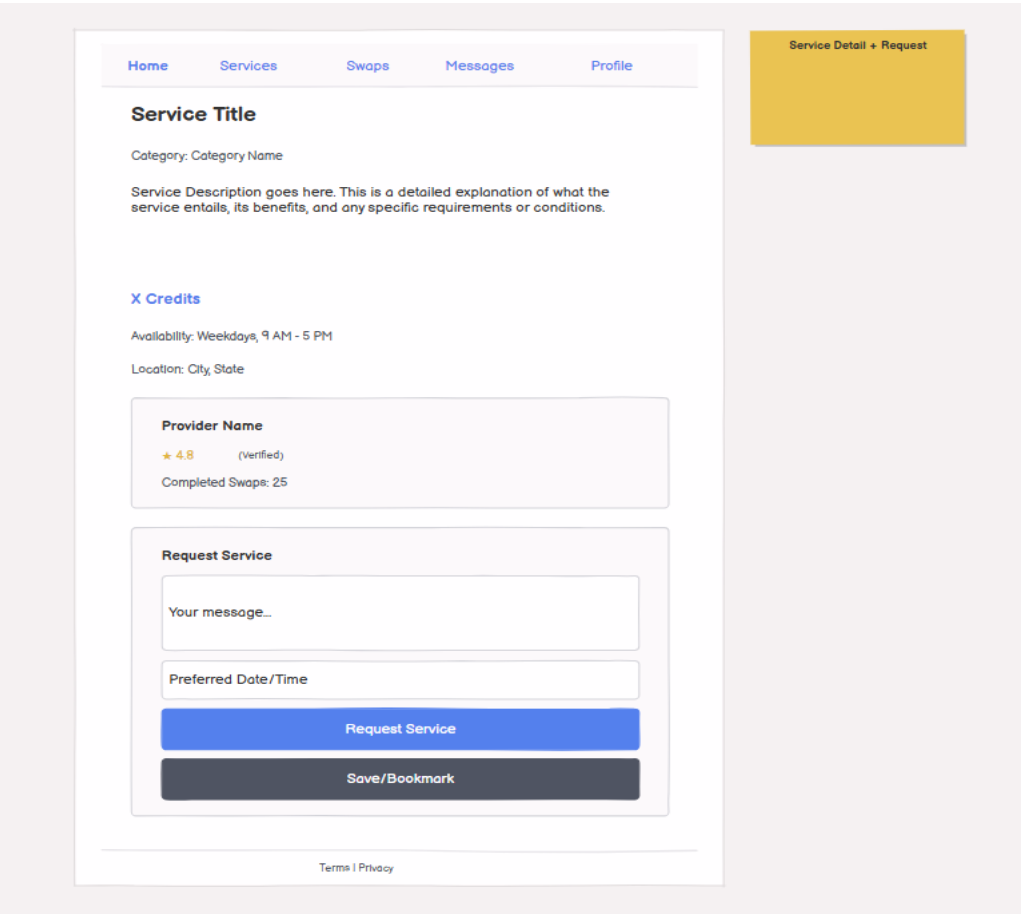
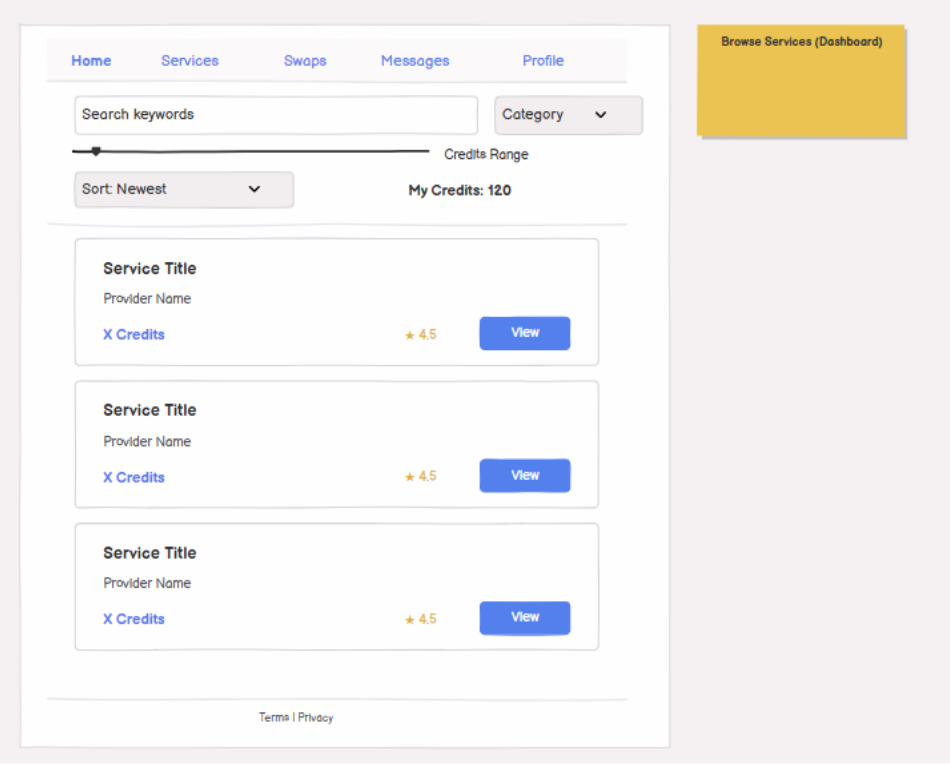
Messaging Interface - allows communication between users during swaps.

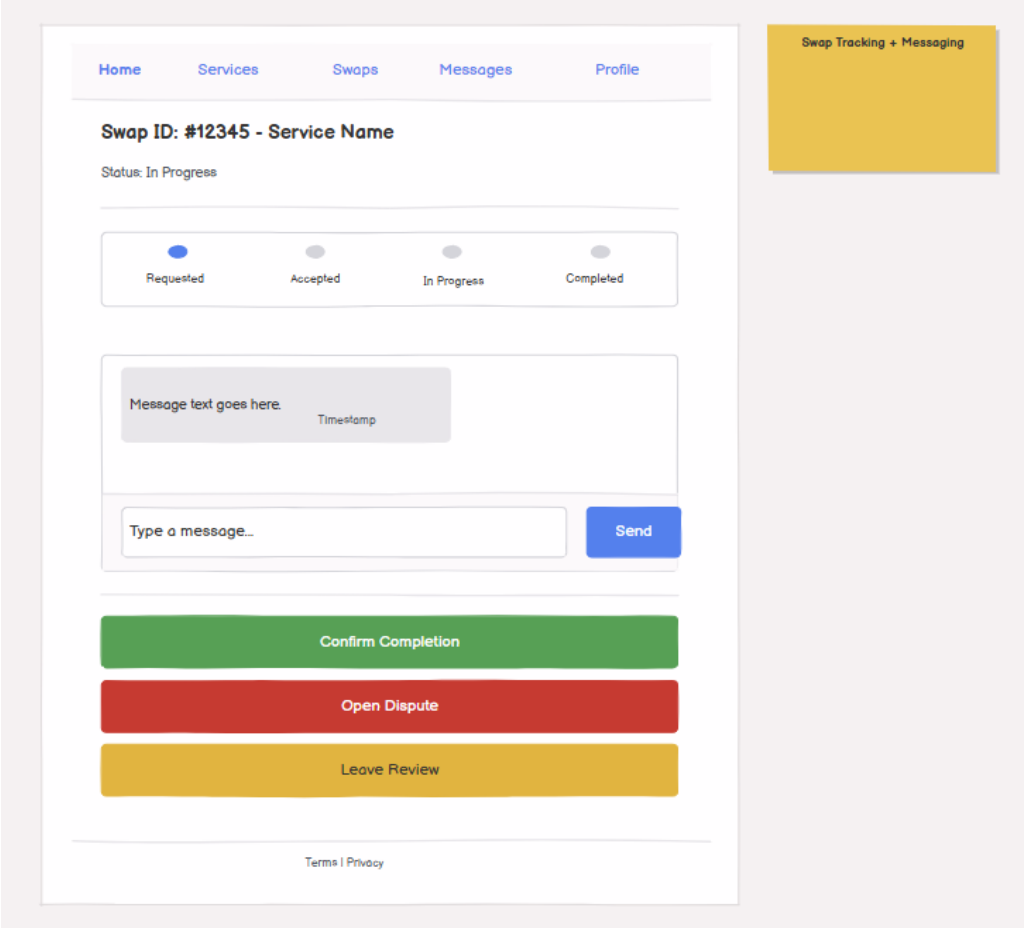
Profile Management Page - allows users to update their skills, availability, and personal information.

Wireframes help designers and developers understand how users will navigate the system.



A wireframe of a login/register page. The page is enclosed in a light gray border. At the top left, there is a placeholder for an app logo (a square with an 'X') and the text "App Name". At the top right, there is a yellow button labeled "Login / Register". Below the app name, there are two input fields: "Email" and "Password". Below these fields are two buttons: a blue "Login" button and a dark gray "Register" button. Below the buttons are two links: "Forgot password" and "Help/Contact". At the bottom left, there is a link "Terms | Privacy".





[Home](#) [Services](#) [Swaps](#) [Messages](#) [Profile](#)



Full Name

★ 4.7 Verified

Active Member

My Credits: 120

+5 added recently

-3 spent recently

About / Skills

Short biography goes here, describing the user's interests and what they offer. Lorem ipsum dolor sit amet, consectetur adipiscing elit.

Skills: Web Design, Graphic Design, Content Writing

Location: City, State

[My Listings](#) [My Swaps](#) [Reviews](#) [Settings](#) [Subscription](#)

Content for the selected tab will appear here. For example, a list of services, active swaps, reviews, or settings forms.

Edit Profile

Manage Subscription

[Terms](#) | [Privacy](#)

Profile Page

3. Revenue Model

Service Swap – Subscription-Based Service Exchange Platform)

Service Swap is designed to generate revenue through a multifaceted monetization strategy focused on ensuring user growth while incentivizing premium features and optional add-ons. The key revenue avenues include subscriptions, advertising, sponsored listings, and optional transaction-based enhancements.

3.1 Subscriptions (Primary Revenue Stream)

Our primary monetization strategy is a tiered subscription model.

- **Free:** plan with limited features intended to increase sign-ups.
- **Premium:** plan with enhanced visibility, unlimited swap requests, and free experience.
- **Pro:** plan offers advanced analytics and lower transaction fees for semi-pros.

3.2 Advertising

- Advertisements displayed only to free-tier users.
- **Sponsored listings:** service providers pay to boost their visibility in search results.

3.3 Optional Transaction-Based Revenue

- Example additional, optional features senders pay to boost their visibility, billing
- ☆ **Profile verification** for a one-time fee.
- ☆ **Boost listing**, a micro-transaction, highlights listings for short time for a small fee.

3.4 Credit System

- Credits as the core of the platform's barter economy, Services rendered and, on referrals.
- The credits can be purchased directly to maintain community-driven exchange system.

Table 3: Revenue Strategy Summary

Revenue Type	Stability	Scalability	Primary?
Subscriptions	High	High	✓ Yes
Sponsored Listings	Medium	High	✓ Secondary
Ads	Medium	High	✓ Secondary
Micro-Transactions	Medium	Medium	✓ Optional
Transaction Fees	Future	High	✓ Expansion

By utilizing subscriptions as the core revenue strategy, complemented by advertising and micro-transactions, Service Swap ensures a sustainable and scalable business model.

8. Conclusion

Progress Report 2 focuses on the system design and architecture of the Service Swap platform. The report explains the architectural structure, backend modules, business logic, database design, and frontend layout of the system.

These technical designs will guide the implementation phase of the project where the system will be developed using PHP and MySQL. The structured architecture ensures that

the platform will be scalable, secure, and easy to maintain as new features are added in the future.